

UAV Compass Variance Simulator

QuickStart Guide

1. Launch the App

Run the following command in your terminal:

```
streamlit run app.py
```

2. Set True Heading

Use the slider to select a true compass heading (0–359°).

3. Select Error Source

Choose from:

- None
- EMI (Electromagnetic Interference)
- Nearby Metal
- Poor Calibration

4. Visualize Compass

Blue arrow = True North

Red arrow = Magnetic (simulated)

5. Check Predicted Drift

See how far off-course the UAV would go over 1 km. Results shown in meters.

6. Observe AI Autopilot Response

Nominal: heading is accurate

Warning: moderate drift detected

RTB: return to base due to high drift

7. Drift Map (Optional)

View a real-time plot showing deviation over 5 km.

8. Recalibrate Compass

Click the button to reset variance and start fresh.

9. Export Logs

Download CSV logs for trend analysis or training use.